

4) a) Variables:

λ = # of customers

$\lambda = 3$ customers/hr

P = Profit = $\$5 + 2y$

H = total heart shaped cookies = $2C$

y = cookie cost

Linearity of expectation

$$E[H] = E[2C] = 2E[C] = 2(24) = 48$$

$$\begin{aligned} 24 \text{ customers per day} &= \lambda \cdot 8 \text{ hr} \\ &= 3 \cdot 8 \\ &= 24 \end{aligned}$$

$$E[P] = E[(5 + 2y)C]$$

$$= (5 + 2y)E[C]$$

$$= 5 + 2y(24)$$

$$= \boxed{120 + 48y} \text{ Expected profit: } E[H] \cdot 5 + 2y$$

which is the profit

generated from the heart
cookies plus the expected
profit